The Warehouse Selection Puzzle

Background

Your company's distribution hub is preparing for a major outbound shipment. The facility holds *exactly one hundred distinct inventory items*. Each item is described by three numerical attributes:

1. A **monetary figure** — the item's assessed value, in whole dollars, ranging from 100 to 1000.
2. A **mass value** — the item's mass in kilograms, an integer between 10 and 100.
3. A **categorical tag** — an integer label from 0 to 9, representing the item's group or supplier.

Selection Guidelines

The procurement and operations teams have provided the following non-negotiable rules:

Rule A (Mass Ceiling): The sum of the *mass values* of all chosen items must **not exceed one metric tonne**. For this puzzle, "one metric tonne" means *exactly* 1000 kilograms; approximations or overages are unacceptable.

Rule B (Category Exclusivity): A shipment must not contain more than *one item* from the same *categorical tag*. If two or more items share the same tag, at most one of them may be selected.

Rule C (Taboo Pairs): Certain unordered pairs of items are deemed *taboo*. These items cannot be shipped together under any circumstances. The complete list of such pairs will be provided separately.

Objective

From the available 100 items, choose a subset whose total *monetary figure* is as large as possible, while respecting all three rules above.

Deliverables

You must produce:

- The identities of the chosen items, including their monetary figure, mass value, and categorical tag.
- The total monetary figure of the chosen set.
- The total mass value of the chosen set.
- A clear demonstration that all three rules (A, B, and C) are satisfied.

Important Note

The terminology in this description is precise. Misreading "mass value" as anything other than kilograms, treating "categorical tag" loosely, or misinterpreting "taboo pairs" will render the solution invalid, even if the total value appears optimal.